

Line integrals:

Parametric
curve $\underline{r}(t)$

When a curve $\underline{r}(t) = g(t)\underline{i} + h(t)\underline{j} + k(t)\underline{k}$, $a \leq t \leq b$, passes through the domain of a function $f(x, y, z)$ in space, the values of f along the curve are given by the Composite Function $f(g(t), h(t), k(t))$. If we integrate this composite with respect to arc length from $t=a$ to $t=b$, we calculate the so called line integral of f along the curve.

If equation of a curve C in the xy -Plane is given as $y = g(x)$, $a \leq x \leq b$ the line integral

$$\int_C f(x, y) ds = \int_a^b f[x, g(x)] \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

If C is given as $x = h(y)$, $c \leq y \leq d$

then

$$\int_C f(x, y) ds = \int_c^d f[h(y), y] \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

If C is given in parametric form

$$x = g(t), \quad y = h(t), \quad t_1 \leq t \leq t_2$$

then

$$\int_C f(x, y) ds = \int_{t_1}^{t_2} f[g(t), h(t)] \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Ex Evaluate $\int_C xy^3 ds$ where

i) C is the segment of the straight line $y = 2x$ from $(-1, -2)$ to $(1, 2)$

ii) C is the circle having parametric equations

$$x = \cos t, \quad y = \sin t \quad 0 \leq t \leq \pi/2$$

Sol

$$\int_C xy^3 ds = \int_{-1}^{+1} x \cdot (2x)^3 \times \sqrt{5} dx$$

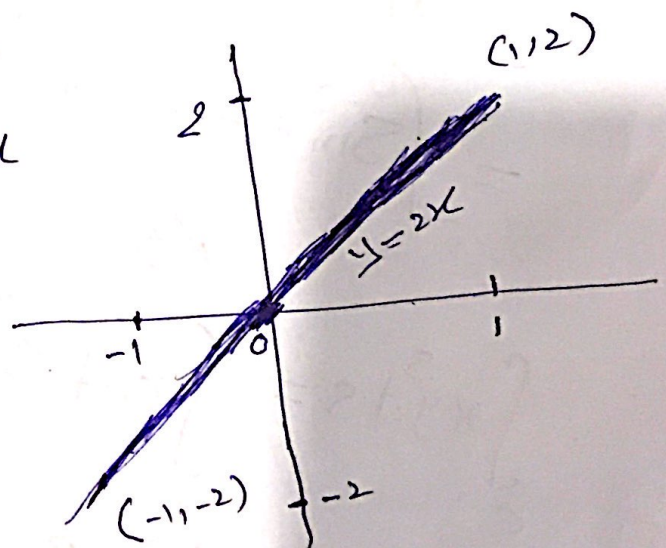
$$= 8\sqrt{5} \int_{-1}^{+1} x^4 dx$$

$$= 8\sqrt{5} \left[\frac{x^5}{5} \right]_{-1}^{+1}$$

$$= 8\sqrt{5} \left(\frac{1}{5} - \left(-\frac{1}{5}\right) \right)$$

$$= 8\sqrt{5} \times \frac{2}{5}$$

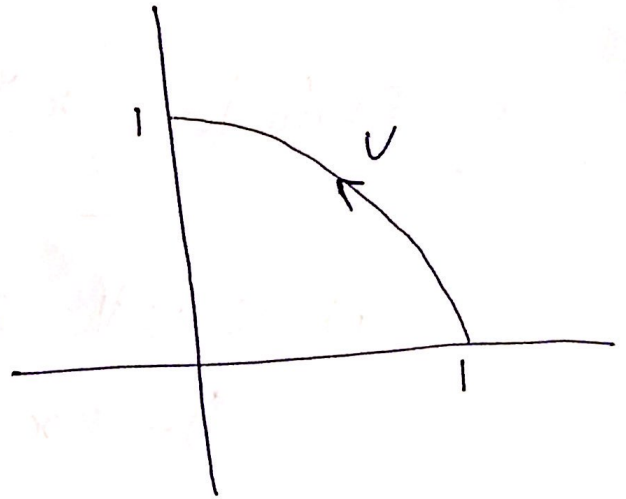
$$= \frac{16}{\sqrt{5}}$$



$$y = 2x$$

$$\frac{dy}{dx} = 2$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + 4} = \sqrt{5}$$



$$\frac{dx}{dt} = -\sin t$$

$$\frac{dy}{dt} = \cos t$$

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$= \sqrt{\sin^2 t + \cos^2 t} = 1$$

$$x = \cos t$$

$$y = \sin t$$

$$x^2 + y^2 = 1^2$$

$$\cos^2 t + \sin^2 t = 1^2$$

$$1^2 = 1$$

$$1 = 1$$

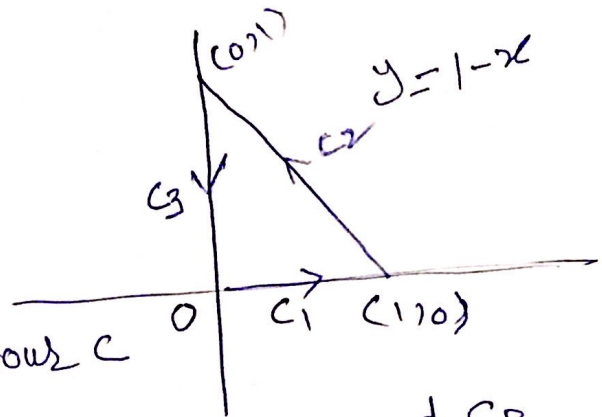
$$\int_C xy^3 ds = \int_0^{\pi/2} \cos t \sin^3 t \times 1 \times dt$$

$$= \int_0^{\pi/2} \sin^3 t \cos t dt$$

$$= \left. \frac{\sin^4 t}{4} \right|_0^{\pi/2} = \frac{1}{4} - 0 = \frac{1}{4}$$

Ex Evaluate $\oint_C (x+y) ds$ where C is 59
the contour of the triangle with
vertices $(0,0)$, $(1,0)$, and $(0,1)$

Sol



In this case, the contour C
consists of the line segments C_1 , C_2 and C_3
We know that the equations of C_1 is $y=0$,
 C_2 is $y=1-x$, and C_3 is $x=0$

$$\text{Thus } \oint_C (x+y) ds = \int_{C_1} (x+y) ds + \int_{C_2} (x+y) ds + \int_{C_3} (x+y) ds$$

For C_1 , $y=0$ Therefore $\frac{dy}{dx}=0$, $\sqrt{1+\left(\frac{dy}{dx}\right)^2}=1$

$$\int_C (x+y) ds = \int_{x=0}^1 (x+0) \cdot 1 dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2}$$

For C_2 $y = 1 - x$ $\frac{dy}{dx} = -1$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + 1} = \sqrt{2}$$

$$\int_{C_2} (x+y) ds = \int_1^0 (x+1-x) \times \sqrt{2} dx$$

$$= \sqrt{2} \int_{x=1}^0 dx = \sqrt{2} x \Big|_1^0 = \sqrt{2} (0-1) = -\sqrt{2}$$

For C_3 $x = 0$, $\frac{dx}{dy} = 0$ $\sqrt{1 + \left(\frac{dx}{dy}\right)^2} = 1$

$$\int_{C_3} (x+y) ds = \int_{y=1}^0 (0+y) \times 1 dy = \frac{y^2}{2} \Big|_1^0 = 0 - \frac{1}{2} = -\frac{1}{2}$$

$$\therefore \oint_C (x+y) ds = \frac{1}{2} - \sqrt{2} - \frac{1}{2} = -\sqrt{2}$$

(Put in eqn (A))

Line Integral along a Space curve, 60

$$x = g(t), y = h(t), z = k(t) \\ a \leq t \leq b$$

$$\int_C f(x, y, z) ds = \int_a^b f[g(t), h(t), k(t)] \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Ex Evaluate $\int_C (xy + z^2) ds$ where C

is the curve defined by

$$x = \cos t, y = \sin t, z = t \\ 0 \leq t \leq \pi$$

$$\begin{aligned} \int_C (xy + z^2) ds &= \int_0^\pi (\cos t \sin t + t^2) \sqrt{2} dt \\ &= \sqrt{2} \left[\frac{\sin^2 t}{2} + \frac{t^3}{3} \right]_0^\pi = \sqrt{2} \left[(0 + \frac{\pi^3}{3}) - (0 + 0) \right] \\ &= \sqrt{2} \pi^3 / 3 \end{aligned}$$

$$\frac{dx}{dt} = -\sin t$$

$$\frac{dy}{dt} = \cos t$$

$$\frac{dz}{dt} = 1$$

$$\begin{aligned} &\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \\ &= \sqrt{2} \end{aligned}$$

Ex Find the line integral of $f(x, y, z) = 2xy + \sqrt{z}$
 over the helix $\underline{r}(t) = \cos t \underline{i} + \sin t \underline{j} + t \underline{k}$
 $0 \leq t \leq \pi$

Sol

$$\int_C (2xy + \sqrt{z}) ds = ?$$

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$$

$$= \sqrt{(-\sin t)^2 + (\cos t)^2 + 1}$$

$$= \sqrt{1+1} = \sqrt{2}$$

$$x = \cos t$$

$$\frac{dx}{dt} = -\sin t$$

$$y = \sin t$$

$$\frac{dy}{dt} = \cos t$$

$$z = t$$

$$\frac{dz}{dt} = 1$$

$$\int_C (2xy + \sqrt{z}) ds = \int_0^\pi (2 \sin t \cos t + t^{1/2}) \sqrt{2} dt$$

$$= \sqrt{2} \left[\sin^2 t + \frac{t^{3/2}}{3/2} \right]_0^\pi$$

$$= \sqrt{2} \left[(0 + \frac{2}{3} \pi^{3/2}) - (0 + 0) \right] = \frac{2\sqrt{2}}{3} \pi^{3/2}$$